# 1. Executive Summary

## System Summary

The Sustainable Finance MEL Platform is intended to be a scalable Monitoring, Evaluation, and Learning platform for the CRDB Sustainable Finance Unit. The current prototype uses TACATDP monitoring as the proof-of-concept use case.

The prototype demonstrates a practical path for CRDB to collect programme/project data, organize it for monitoring, and present key indicators through the portal while a more scalable analytics and governance layer is planned.

## Problem Being Addressed

Sustainable Finance Unit programmes and operational initiatives require reliable data on activities, beneficiaries, resources, costs, efficiency gains, environmental outcomes, and institutional impact. Manual or loosely structured collection creates recurring problems:

- inconsistent data capture;
- difficult follow-up across the same beneficiary, project, facility, or operational process;
- weak traceability between baseline, intervention, and later results;
- slow reporting;
- limited visibility for managers and reviewers;
- high effort to prepare data for dashboards and analysis.

Sustainable Finance MEL Platform addresses this by creating a structured digital collection and monitoring prototype.

## Prototype Purpose

The TACATDP proof of concept is intended to prove that:

- beneficiary and programme baseline data can be collected through a guided digital form;
- the data can be structured into monitoring-ready records;
- beneficiaries can be modeled as linkable entities rather than only one-off form responses;
- basic KPIs can be visualized directly in the portal without waiting for Power BI permissions;
- the same foundation can evolve into a configurable Sustainable Finance MEL Platform if accepted.

## Current Prototype Capability

The current prototype supports baseline data collection using a Power Platform/Power Pages form derived from the TACATDP XLSForm. The implementation includes a structured form flow, schema artifacts, reference data, validation planning, and deployment/runbook documentation.

The immediate prototype revision adds two important acceptance-facing capabilities:

1. lightweight beneficiary entity modeling;
2. direct portal KPI visualisation for key monitoring indicators.

## Intended Users

| User group | Main need |
|---|---|
| Beneficiaries/respondents | Provide baseline information through a structured form. |
| Enumerators/data collectors | Capture complete, valid data with guided sections and validation feedback. |
| MEL officers/reviewers | Review beneficiary submissions and monitor coverage/progress. |
| Project managers | See headline KPIs and understand programme reach. |
| Sustainable Finance Unit managers | Monitor multiple programmes/projects and compare evidence across initiatives. |
| System administrators | Configure, deploy, troubleshoot, and maintain the prototype. |
| Technical developers | Extend the prototype into a scalable product. |

## Business and Programme Value

The prototype adds value by:

- reducing reliance on manual spreadsheets and unstructured data collection;
- improving consistency of beneficiary baseline records;
- making beneficiary and location coverage easier to monitor;
- preparing data for follow-up, endline, and impact analysis;
- giving reviewers visible KPIs during assessment;
- creating a foundation for a governed Sustainable Finance MEL Platform that can support multiple use cases.

Beyond the TACATDP proof of concept, the same platform capability can support a broader Sustainable Finance Unit operating scope, including configurable programmes, projects, facilities, operational processes, resources, activities, indicators, evidence, and institutional impact monitoring.

## Prototype Boundary

Treat the current TACATDP-based prototype as a proof of concept and demonstration system. It does not claim full production readiness or full coverage of all future Sustainable Finance Unit use cases.

The prototype focuses on:

- TACATDP baseline collection;
- beneficiary-linked data structure;
- portal-level KPI visibility;
- documented deployment and handover.

Production readiness items such as enterprise master-data governance, full Power BI semantic modeling, formal support, high-availability operations, and integration with core CRDB systems belong to the future product roadmap.

## Future Product Direction

If accepted, Sustainable Finance MEL Platform should evolve into a scalable MEL platform with:

- central beneficiary registry;
- configurable programme/project templates;
- facility/process/resource monitoring entities;
- baseline, follow-up, and endline tracking;
- stronger access control and audit trails;
- governed reporting and analytics;
- Power BI dashboards;
- data quality review workflows;
- formal deployment, support, and release management.

## Acceptance Message

The prototype should be assessed on whether it demonstrates a credible digital MEL workflow: structured beneficiary baseline collection, monitoring-ready data design, portal KPI visibility, and a clear roadmap toward a robust production product.

# 2. Prototype Scope

## Prototype Summary

The current Sustainable Finance MEL Platform prototype uses TACATDP monitoring as a proof-of-concept use case for digital Monitoring, Evaluation, and Learning. The current implementation is based on the TACATDP impact evaluation form and Power Platform deployment work.

The prototype demonstrates how programme/project data can be collected digitally, structured for monitoring, and surfaced through simple portal-level indicators. TACATDP is the first demonstration case; it is not the full boundary of the intended platform.

## In Scope

The prototype scope includes:

- baseline data collection from beneficiaries through a deployed portal form;
- structured form sections derived from the TACATDP XLSForm;
- reference data for locations, CRDB branches, and coded choices;
- beneficiary-linked submission records;
- lightweight beneficiary entity modeling;
- basic portal KPI visualisation for immediate monitoring;
- documentation for users, administrators, reviewers, and future developers.

The prototype should also document how the same platform pattern can later support other Sustainable Finance Unit use cases, even if they are not implemented in the current prototype.

## Beneficiary Entity Model

The prototype should model beneficiaries as identifiable records linked to baseline submissions.

At prototype level, the beneficiary model should remain lightweight:

| Entity | Purpose |
|---|---|
| Beneficiary | Identifies the person, household, group, or enterprise being monitored. |
| Submission / Baseline Assessment | Stores one baseline form submission linked to a beneficiary. |
| Reference Data | Supports location, branch, value-chain, and coded-choice filtering. |

Minimum beneficiary fields should include:

- beneficiary identifier;
- beneficiary name or display label where allowed;
- beneficiary type or grouping where applicable;
- sex, age category, youth/women/social inclusion attributes where collected;
- region, district, ward, and village;
- value chain or intervention category;
- source submission metadata.

This supports deduplication, follow-up monitoring, and future baseline-to-endline comparisons.

## Portal KPI Visualisation

Because Power BI access and permissions may take time, the prototype should include basic portal-level visualisation for key indicators.

Recommended prototype KPIs:

- total baseline submissions;
- total beneficiaries captured;
- beneficiaries by region/district;
- beneficiaries by value chain;
- gender/youth/social inclusion summary where data exists;
- form completion or submission status;
- baseline production, yield, income, water, or GHG indicators where enough data exists.

The dashboard should be documented as direct portal visualisation. Power BI should be documented as a future analytics enhancement, not as a dependency for prototype acceptance.

## Out of Scope for Prototype

The prototype does not yet claim:

- full enterprise beneficiary master-data governance;
- full Power BI semantic model and dashboard deployment;
- production-grade integration with CRDB core systems;
- automated data quality workflow and approval pipeline;
- complete offline-first mobile field collection;

- full production support, backup, disaster recovery, and monitoring.

These belong to the scalable product vision.

The prototype also does not yet implement broader non-TACATDP operating use cases. Those should be documented as future configurable programme/project, facility, process, resource, indicator, and evidence capabilities rather than as specific named initiatives.

## Acceptance Position

The prototype is acceptable if it demonstrates:

- working baseline data collection;
- clear beneficiary-linked data structure;
- visible monitoring indicators on the portal;
- documented deployment and administration process;
- known limitations and realistic roadmap toward production scale.

# 3. Requirements Specification

## Purpose

This specification defines the prototype requirements for the Sustainable Finance MEL Platform and separates them from future-product requirements.

The current prototype uses TACATDP monitoring as the proof-of-concept use case. Future-product requirements must support a configurable platform for multiple Sustainable Finance Unit programmes/projects.

## Scope Classification

| Scope | Meaning |
|---|---|
| Prototype current | Already represented in the current prototype artifacts or deployed baseline collection flow. |
| Prototype revision | Agreed addition before final prototype documentation/delivery. |
| Future product | Required for a scalable, robust production-grade Sustainable Finance MEL Platform. |

## Prototype Functional Requirements

| ID | Requirement | Scope | Priority | Acceptance Criteria |
|---|---|---|---|---|
| PR-FR-01 | The system must provide a guided baseline data collection form derived from the TACATDP impact evaluation XLSForm. | Prototype current | Must | Users can access the form and move through structured sections. |
| PR-FR-02 | The system must preserve relevant baseline survey sections, labels, choices, required rules, and validation intent. | Prototype current | Must | Form sections and validation behavior can be traced to the XLSForm inventory and validation/save map. |
| PR-FR-03 | The system must store submitted data in structured lists/entities rather than an unstructured document. | Prototype current | Must | Submission data maps to documented schema/list artifacts. |
| | The system must support reference data for location, | | Must | Region, district, ward, village, branch, and coded-choice |

| ID | Requirement | Scope | Priority | Acceptance Criteria |
|---|---|---|---|---|
| PR-FR-04 | CRDB branch, and coded choices. | Prototype current | | references are represented in schema/import artifacts. |
| PR-FR-05 | The system must model beneficiaries as identifiable records linked to baseline submissions. | Prototype revision | Must | A beneficiary identifier or equivalent link exists between beneficiary records and submitted baseline assessments. |
| PR-FR-06 | The system must support beneficiary segmentation for monitoring where data is collected. | Prototype revision | Should | Users/reviewers can group beneficiary counts by location, value chain, sex, youth/women/social inclusion attributes, or other available fields. |
| PR-FR-07 | The system must show basic KPI visualisation directly on the portal. | Prototype revision | Must | Portal page displays KPI cards/ charts for key indicators without requiring Power BI. |
| PR-FR-08 | The system must distinguish draft, submitted, failed, and reviewed states where supported by the data model. | Prototype current/ revision | Should | Submission status is visible or represented in the data model. |
| PR-FR-09 | The system must provide user-facing validation feedback for required and invalid visible fields. | Prototype current/ revision | Must | Invalid visible fields block continuation/submission and show useful messages. |
| PR-FR-10 | The system must provide documentation for users and administrators. | Prototype revision | Must | User manual and deployment/admin guide exist under the submission documentation pack. |
| PR-FR-11 | The documentation must position TACATDP as the first proof-of-concept use case, not as the full platform boundary. | Prototype revision | Must | Submission documents explain the multi-programme Sustainable Finance Unit platform direction. |

## Portal KPI Requirements

| ID | KPI Requirement | Scope | Acceptance Criteria |
|---|---|---|---|
| KPI-01 | Show total baseline submissions. | Prototype revision | Dashboard displays a count based on available submission records or verified demo data. |
| KPI-02 | Show total beneficiaries captured. | Prototype revision | |

| ID | KPI Requirement | Scope | Acceptance Criteria |
|---|---|---|---|
| | | | Dashboard displays total beneficiaries or beneficiary counts from the current data model. |
| KPI-03 | Show geographic coverage. | Prototype revision | Dashboard breaks data down by region/ district where location fields exist. |
| KPI-04 | Show value-chain coverage. | Prototype revision | Dashboard summarizes beneficiaries or submissions by value chain where fields exist. |
| KPI-05 | Show social inclusion indicators where data exists. | Prototype revision | Dashboard summarizes gender, youth, women, or other inclusion indicators when captured. |
| KPI-06 | Clearly label demo or sample data if live data is unavailable. | Prototype revision | Reviewer can distinguish real submitted data from seeded/demo values. |

## Beneficiary Modeling Requirements

| ID | Requirement | Scope | Acceptance Criteria |
|---|---|---|---|
| BEN-01 | The prototype must represent a beneficiary separately from a one-time form submission. | Prototype revision | Beneficiary concept is documented and represented in the data model or implementation plan. |
| BEN-02 | Each baseline submission must link to a beneficiary identifier where feasible. | Prototype revision | Submission record can be associated with a beneficiary. |
| BEN-03 | Beneficiary data must support future follow-up and endline comparisons. | Prototype revision/future | Data model does not prevent multiple submissions over time for the same beneficiary. |
| BEN-04 | The future product must support central beneficiary master data and deduplication. | Future product | Roadmap includes registry governance, deduplication, and data ownership. |

## Non-Functional Requirements

| ID | Requirement | Scope | Acceptance Criteria |
|---|---|---|---|
| NFR-01 | Documentation and source artifacts must be version controlled. | Prototype current | Repository contains docs, schema, scripts, and app artifacts. |
| NFR-02 | | | |

| ID | Requirement | Scope | Acceptance Criteria |
| --- | --- | --- | --- |
| | The prototype must avoid storing secrets or credentials in source files. | Prototype current | No secrets are committed in documentation or scripts. |
| NFR-03 | Deployment and authentication steps must be documented with known tenant/ profile constraints. | Prototype revision | Admin guide records PAC and environment notes. |
| NFR-04 | The prototype must not depend on Power BI permissions for basic assessment visibility. | Prototype revision | Portal KPI dashboard provides immediate monitoring visibility. |
| NFR-05 | The future product must define production-grade security, audit, backup, and support requirements. | Future product | Future-product vision and roadmap include production readiness gates. |
| NFR-06 | The future product must not be constrained to Power Pages if enterprise requirements require a more robust application architecture. | Future product | Future vision documents target frontend, backend API, DBMS/data platform, integration, hosting, and operations review. |

## Data Requirements

The prototype data model should support:

- parent submission records;
- beneficiary records or beneficiary-linked summary records;
- section-based baseline data;
- multi-select answer rows where needed for analytics;
- production cost line rows where repeated cost data exists;
- reference data for locations, branches, and choices;
- status and timestamp metadata.

Existing TACATDP list/schema names may remain unchanged for compatibility with generated artifacts and deployed components.

## User Roles

| Role | Required access |
| --- | --- |
| Beneficiary/respondent | Submit baseline data where portal access model allows. |
| Enumerator | Capture and save baseline data. |

| Role | Required access |
| --- | --- |
| MEL reviewer | Review submissions and dashboard indicators. |
| Administrator | Configure portal, data sources, users, and deployment settings. |
| Developer | Maintain source, schema, scripts, and documentation. |

## Out of Scope for Prototype

The prototype does not include:

- guaranteed production SLA;
- full Power BI dashboard integration;
- enterprise master-data management;
- integration with CRDB core banking or customer systems;
- complete offline synchronization;
- automated approval workflow unless explicitly implemented;
- formal disaster recovery implementation.

## Future Product Requirements

If accepted, the scalable Sustainable Finance MEL Platform should include:

- configurable programme/project setup;
- governed beneficiary registry;
- configurable entities for facilities, processes, resources, activities, indicators, and evidence;
- reviewed frontend/application architecture beyond the Power Pages proof of concept;
- backend API/service layer;
- enterprise DBMS or approved governed data platform;
- advanced analytics and Power BI dashboards;
- data quality workflow;
- role-based permissions and audit trails;
- environment-separated ALM process;
- backup, monitoring, incident response, and support model;
- integration strategy for approved enterprise systems.

Future use cases should be expressed as configurable Sustainable Finance Unit operating domains, including programmes, projects, facilities, business processes, resources, activities, indicators, evidence, and institutional impact monitoring. Specific initiative examples should remain implied unless approved for inclusion in the final submission.

## Documentation Acceptance Criteria

The documentation pack is acceptable when:

- it explains the prototype clearly for technical and non-technical readers;
- it does not confuse current prototype capability with future product vision;
- beneficiary modeling and portal KPI visualisation are included;
- requirements map to architecture, testing, and traceability artifacts;
- known limitations and future work are explicit.

# 4. Current Architecture

## Architecture Summary

The current Sustainable Finance MEL Platform prototype uses Microsoft Power Platform to collect TACATDP beneficiary baseline data through a guided digital form and organize the captured data into monitoring-ready records.

The active prototype architecture is Power Pages plus Dataverse. Power Pages provides the web portal/user interface, while Dataverse provides the current application data store for project/form configuration, assignments, submissions, and related monitoring records.

The current implementation is derived from the TACATDP XLSForm and related generated artifacts. TACATDP names remain in some schema/list/form artifacts for compatibility and traceability. Treat TACATDP as the first configurable programme/project use case, not as the full platform boundary.

## Current Prototype Architecture

```
 User / Enumerator
   -> Power Pages portal / hosted form runtime
   -> Power Pages Web API
   -> Dataverse tables/entities
   -> Project, form, assignment, submission, beneficiary, reference, and
evidence records
   -> Portal KPI dashboard
```

## Main Components

| Component | Purpose | Current status |
| --- | --- | --- |
| Power Pages portal | Provides browser-based access to the baseline form, authenticated user experience, and planned dashboard. | Deployed prototype site exists. |
| Baseline form | Captures beneficiary baseline data from the TACATDP-derived form. | Current prototype capability. |
| Dataverse | Current prototype data store for Power Pages configuration and application records. | Active delivery path. |
| | | Active delivery path. |

| Component | Purpose | Current status |
|---|---|---|
| Power Pages Web API | Browser-to-Dataverse write/read channel for authenticated portal users. | |
| Submission record | Parent record for a captured baseline assessment. | Represented in Dataverse/schema/design artifacts. |
| Beneficiary model | Identifies beneficiaries separately from one-off submissions. | Agreed prototype revision. |
| Section/form response records | Store detailed responses by form section or structured payload. | Represented through generated schema and Dataverse design artifacts. |
| Reference data | Supports locations, branches, and choice lists. | Represented through import templates, schema artifacts, and Dataverse-ready design. |
| Portal KPI dashboard | Shows headline indicators without waiting for Power BI. | Agreed prototype revision. |
| Power BI analytics | Future governed analytics layer. | Future product scope. |
| Programme/project configuration | Allows the platform to support multiple Sustainable Finance Unit use cases. | Future product scope. |

## Data Model Direction

The prototype should avoid treating a submitted form as the only data object. In Dataverse, it should separate:

- programme/project;
- beneficiary identity;
- baseline submission;
- form section responses;
- repeated/multi-select answer rows;
- reference data;
- calculated or aggregated KPI values.

This keeps the prototype useful for monitoring and creates a cleaner migration path toward a production MEL platform.

The future data model should also support non-beneficiary use cases by allowing configurable monitored entities for facilities, operational processes, resources, activities, indicators, results, and evidence attachments.

## Prototype Versus Enterprise Architecture

Power Pages and Dataverse are appropriate for the proof-of-concept because they provide fast delivery, Microsoft 365/Power Platform alignment, authentication integration, and a manageable route for demonstrating data collection and portal KPIs.

For an enterprise-grade scalable product, the architecture should be reviewed beyond the current Power Pages implementation. The future platform may retain some Microsoft ecosystem components where they fit, but it should not be constrained to Power Pages as the long-term application shell.

The target architecture should evaluate:

- a dedicated web application frontend for richer UX, performance, and maintainability;
- a backend API/service layer for business rules, workflow, integration, and audit;
- an enterprise DBMS or governed data platform for transactional and analytical data;
- Dataverse where it remains appropriate for Microsoft-native workflow, configuration, or rapid business app delivery;
- an analytics layer such as Power BI or another approved BI stack;
- identity, access control, audit, monitoring, backup, and release management as first-class architecture concerns.

## Beneficiary-Linked Model

```
Beneficiary
  -> Baseline Submission
      -> Profile / demographics section
      -> Agriculture section
      -> Resource efficiency section
      -> Social inclusion section
      -> Beneficiary quantification sections
      -> Safeguards and climate section
      -> Insurance / guarantee section
      -> GHG, water, yield, income sections
      -> Production cost lines
```

At prototype level, beneficiary modeling may be lightweight. At future-product level, it should become a governed beneficiary registry with deduplication, identity rules, audit, and lifecycle management.

## Reference Data

The source form includes large and filtered choice lists. Location and branch choices should use reference data rather than hard-coded dropdown values.

Important reference categories include:

- region;
- district;
- ward;
- village;
- CRDB branch;
- coded form choices.

Large reference lists should be filtered using stable indexed keys where supported by the selected data source.

## Portal KPI Dashboard

The prototype dashboard should display key indicators directly in the portal:

- total baseline submissions;
- total beneficiaries captured;
- beneficiaries by location;
- beneficiaries by value chain;
- social inclusion indicators where available;
- submission status or completion counts.

This is a prototype analytics layer. Power BI remains the recommended future-product analytics layer once permissions, licensing, ownership, and governance are approved.

## Authentication and Access

Authentication and deployment are governed by the target Power Platform tenant/environment.

Known operational lessons:

- PAC authentication should use verified environment IDs where possible.
- Do not assume Azure service principal or managed identity authentication unless explicitly approved.
- CRDB deployment work historically used device-code authentication with the delegated `dmuroba@crdb.co.tz` profile.
- Mshirika access must be verified against the correct tenant account and Power Pages environment.

## Current Architecture Limitations

The current prototype does not yet provide:

- full production master-data governance;
- complete Power BI analytics integration;
- enterprise-grade audit and role model;
- formal backup, monitoring, and incident management;
- integration with CRDB enterprise systems;
- full offline-first data collection.

These are future-product architecture requirements.

## Architecture Acceptance Criteria

The current architecture is acceptable for prototype submission if:

- baseline collection is demonstrable;
- data is structured and beneficiary-linked;
- portal KPIs are visible or clearly planned for the prototype revision;
- deployment/authentication constraints are documented;
- production-grade gaps are explicitly assigned to the future product roadmap.

# 5. User Manual

## Purpose

This manual explains how users interact with the Sustainable Finance MEL Platform prototype. The current prototype uses TACATDP monitoring as the proof-of-concept use case.

## User Groups

| User | Main actions |
|---|---|
| Beneficiary/respondent | Provide baseline information where self-service access is enabled. |
| Enumerator/data collector | Capture beneficiary baseline data through the portal form. |
| MEL reviewer | Review submitted records and monitor basic KPI indicators. |
| Administrator | Manage access, deployment configuration, and troubleshooting. |

## Accessing the Prototype

1. Open the deployed Power Pages portal URL provided by the administrator.
2. Sign in using the approved work/school account.
3. If access is denied, contact the administrator. Access may require both site visibility permission and application-level role/assignment configuration.

## Baseline Data Collection Workflow

1. Sign in to the portal.
2. Open the assigned TACATDP baseline monitoring form.
3. Complete each visible section.
4. Review validation messages before continuing.
5. Save or submit the form according to the available action.
6. Confirm that the submission is recorded.

## Beneficiary Information

The prototype should support beneficiary-linked records. This means a submitted baseline assessment should be associated with a beneficiary identity or beneficiary summary record instead of existing only as a one-off form response.

This supports later follow-up, monitoring, and comparison across reporting periods.

## Portal KPI Dashboard

The prototype should expose basic monitoring indicators directly in the portal. These may include:

- total baseline submissions;
- total beneficiaries captured;
- beneficiary coverage by location;
- beneficiary coverage by value chain;
- social inclusion indicators where data exists;
- submission status or completion counts.

If live data is not yet available, any demo or sample dashboard data must be clearly labelled.

## Common Access Issues

| Issue | Likely cause | Action |
|---|---|---|
| Sign-in succeeds but portal access is denied | User lacks Power Pages site visibility or app role/assignment | Ask administrator to verify site visibility, contact, web role, and active assignment. |
| User sees no assigned form/project | Assignment is missing or inactive | Ask administrator to verify active assignment records in Dataverse. |
| Form does not submit | Required/visible fields are missing or invalid | Review validation messages and complete required fields. |
| Dashboard looks empty | No submitted records or data permissions issue | Confirm submissions exist and reviewer permissions are configured. |

## User Responsibilities

- Enter accurate data.
- Do not share account credentials.

- Report validation, access, or data issues to the administrator.
- Do not treat the prototype as production-ready unless formally approved.

## Prototype Limitation Notice

The current system is a proof-of-concept prototype. It demonstrates the MEL workflow, data structure, and portal-level visibility, but full production readiness requires additional security, governance, analytics, support, and architecture review.

# 6. Testing and Validation Report

## Purpose

This report defines how the Sustainable Finance MEL Platform prototype should be validated for submission. It focuses on the TACATDP proof-of-concept workflow and the agreed prototype revisions.

## Validation Scope

| Area | Validation target |
|---|---|
| Portal access | Approved users can reach the Power Pages site. |
| Authentication | Users sign in with approved tenant accounts. |
| Form workflow | Users can open and complete the baseline form. |
| Validation | Required and invalid visible fields are handled correctly. |
| Dataverse persistence | Submitted records are stored in the current data layer. |
| Beneficiary linkage | Submissions can be associated with beneficiary records or beneficiary summary identifiers. |
| KPI dashboard | Portal displays agreed key indicators. |
| Documentation | Prototype and future-product claims are separated. |

## Test Cases

| ID | Test | Expected Result | Status |
|---|---|---|---|
| TST-01 | Open portal URL as an approved user | Portal loads successfully after sign-in | To verify |
| TST-02 | Open assigned TACATDP baseline form | Form is visible to the assigned user | To verify |
| TST-03 | Attempt to continue with missing required visible fields | User sees validation errors and cannot continue/submit | To verify |
| TST-04 | Complete required visible fields | User can proceed through the form | To verify |
| TST-05 | Submit a valid baseline record | Submission is stored in Dataverse | To verify |

| ID | Test | Expected Result | Status |
| --- | --- | --- | --- |
| TST-06 | Check beneficiary linkage | Submission references a beneficiary identity or beneficiary summary model | To verify |
| TST-07 | Open portal KPI dashboard | KPI cards/charts render without Power BI dependency | To verify |
| TST-08 | Verify dashboard data source | Dashboard values use live records or clearly labelled demo data | To verify |
| TST-09 | Test non-admin user access | User has site visibility, portal role, and active assignment | To verify |
| TST-10 | Review documentation pack | Docs separate prototype capability from future scalable product vision | In progress |

## Evidence Required

For final submission, collect:

- screenshots of portal sign-in or landing page;
- screenshots of the baseline form;
- screenshots of validation behavior;
- screenshots of a successful submission or record confirmation;
- screenshots of the portal KPI dashboard;
- exported or inspected Dataverse records where allowed;
- notes on any failed tests and workarounds.

## Known Validation Constraints

- Power BI integration is not required for prototype acceptance.
- Portal KPI visualisation can be implemented directly on the portal.
- User access depends on tenant, Power Pages visibility, contact, web role, and assignment configuration.
- Production readiness is not claimed until architecture, security, governance, and support checks are complete.

## Result Summary

Current status: documentation validation is in progress. Runtime validation should be completed after the final prototype revision for beneficiary modeling and portal KPI visualisation.

# 7. Deployment and Administration Guide

## Purpose

This guide explains the deployment and administration considerations for the Sustainable Finance MEL Platform prototype. The current proof of concept uses TACATDP monitoring, Power Pages, and Dataverse.

## Current Prototype Architecture

```
Power Pages portal
  -> Power Pages Web API
  -> Dataverse tables/entities
  -> Portal KPI dashboard
```

## Administrator Responsibilities

- Confirm the correct Power Platform environment.
- Verify the Power Pages site is connected to the expected Dataverse environment.
- Manage user access, site visibility, portal roles, and active assignments.
- Verify deployment state after changes.
- Avoid committing or exposing secrets.
- Document all environment-specific steps.

## Environment Verification

Use Power Platform CLI where available:

```
 pac auth list
pac pages list
```

When PAC access fails, prefer verified Maker Portal environment IDs over guessed Dataverse organization URLs.

## Authentication Notes

- Do not assume Azure service principal or managed identity authentication unless explicitly approved and verified.
- Historical CRDB deployment work used device-code authentication with the delegated `dmuroba@crdb.co.tz` profile.
- Mshirika access must be verified against the correct tenant account and Power Pages environment.
- If sign-in succeeds but resource access fails, check tenant/environment permissions before diagnosing application code.

## User Access Checklist

For a non-admin user:

1. Confirm the user exists or can authenticate in the relevant tenant.
2. Confirm Power Pages site visibility access where the site is private.
3. Confirm the Power Pages contact exists.
4. Confirm the correct web role is linked.
5. Confirm the relevant form/project assignment exists.
6. Confirm assignment lifecycle/status is active.
7. Ask the user to sign out/in and retest.

## Deployment Verification Checklist

After deployment or configuration changes:

1. Confirm the portal loads.
2. Confirm the correct site/environment is being inspected.
3. Confirm latest web assets are active.
4. Confirm Dataverse records needed by the portal exist.
5. Confirm assigned users can see their form/project.
6. Submit or inspect a test record where approved.
7. Verify portal KPI dashboard renders expected values.
8. Record screenshots and command output for the testing report.

## Safety Rules

- Do not deploy to production without explicit approval.
- Do not print or commit secrets, tokens, `.env` values, authorization headers, or private keys.
- Do not commit raw Power Pages exports containing secret-bearing site settings.
- Do not modify production data without explicit approval.
- Do not rename deployed Dataverse tables/columns/forms without migration review.

## Troubleshooting

| Symptom | Check |
|---|---|
| User signs in but has no access | Site visibility, contact, web role, assignment, tenant permissions |
| User sees no project/form | Active assignment and lifecycle status |
| Portal still shows old UI | Cache purge, localized page content, asset references |
| PAC cannot list site | Correct auth profile, environment ID, tenant access |
| Dataverse write fails | Web API permissions, table permissions, choice values, required fields |

## Future Production Administration

The production platform should add formal:

- environment strategy;
- release management;
- backup and recovery;
- monitoring and observability;
- incident response;
- access review;
- data retention and privacy controls.

# 8. Known Limitations

## Purpose

This document records known prototype limitations so reviewers can distinguish between current proof-of-concept capability and future production requirements.

## Prototype Limitations

| Limitation | Impact | Future Resolution |
| --- | --- | --- |
| TACATDP is the only implemented proof-of-concept use case | The platform concept is demonstrated through one programme workflow | Add configurable programme/project templates and generic monitored entities |
| Beneficiary model is lightweight | It supports prototype tracking but not full master-data governance | Implement central beneficiary registry, deduplication, lifecycle, audit, and ownership rules |
| Portal KPI dashboard is basic | Immediate monitoring is possible, but analytics depth is limited | Add governed analytics layer and semantic reporting model |
| Power BI is not yet connected | Advanced dashboards are not part of the current prototype | Integrate Power BI or approved BI tool after permission, licensing, and governance approval |
| Power Pages is the prototype shell | It may not satisfy all long-term enterprise UX, integration, or scalability requirements | Review target frontend/application architecture before production |
| Dataverse is the current prototype data store | It must be reviewed against enterprise scale, integration, and governance needs | Decide final DBMS/data platform through architecture review |
| Access setup depends on tenant/environment configuration | User onboarding may fail if site visibility, roles, or assignments are incomplete | Formalize access provisioning and support procedures |
| Offline-first collection is not complete | Field collection may depend on connectivity | Define offline requirements and synchronization model |
| Production operations are not formalized | No full SLA, backup, monitoring, or incident process is claimed | Add production operations plan before rollout |

## Claims Not Made

The current prototype does not claim:

- full production readiness;
- full enterprise scalability;
- complete multi-programme configuration;
- complete Power BI implementation;
- complete offline synchronization;
- integration with internal enterprise systems;
- formal support or SLA coverage.

## Submission Position

The prototype should be evaluated as a proof of concept showing:

- digital baseline collection;
- beneficiary-linked monitoring structure;
- Dataverse-backed prototype data model;
- portal-level KPI visibility;
- clear path toward a scalable Sustainable Finance MEL Platform.

# 9. Future Product Vision

## Vision

The Sustainable Finance MEL Platform should evolve from a TACATDP proof-of-concept baseline data collection portal into a scalable Monitoring, Evaluation, and Learning platform for Sustainable Finance Unit programmes/projects, operational impact monitoring, performance reporting, and evidence-based decision support.

The future product should preserve the working prototype lessons while strengthening architecture, security, configurability, data governance, analytics, and operations.

## Target Capabilities

The scalable Sustainable Finance MEL Platform should support:

- configurable programme/project workspaces;
- central beneficiary registry and deduplication;
- configurable monitored entities such as beneficiaries, facilities, operational processes, resources, activities, indicators, results, and evidence records;
- baseline, follow-up, and endline data collection;
- intervention tracking by value chain, location, and beneficiary segment;
- data quality review and approval workflows;
- role-based access for enumerators, reviewers, administrators, and decision makers;
- Power BI or equivalent analytics with governed datasets;
- audit trails for important data changes;
- environment-aware deployment and release management;
- documented support and maintenance procedures.

## Broader Platform Coverage

| Coverage area | What the platform should support |
| --- | --- |
| TACATDP monitoring | Beneficiaries, interventions, baseline/follow-up data, value chains, locations, and impact indicators. |
| Programmes and projects | Configurable activities, indicators, evidence, targets, results, and reporting templates. |

| Coverage area | What the platform should support |
|---|---|
| Facilities and operations | Configurable monitoring of facilities, resources, processes, efficiency indicators, responsible teams, and improvement actions. |
| Institutional impact | Evidence capture for operational improvements, efficiency gains, sustainability outcomes, and management reporting. |

## Target Architecture Direction

The production-grade product should move beyond the proof-of-concept dependency on Power Pages as the main application shell. Power Pages is useful for the current prototype, but the enterprise platform should be assessed against a more robust target architecture.

The production-grade product should move toward:

- a dedicated web application frontend where richer UX, scalability, maintainability, and integration requirements exceed Power Pages capability;
- a backend API/service layer for workflow, validation, business rules, integrations, audit, and secure data access;
- a governed enterprise DBMS or approved data platform for transactional records;
- Dataverse where it remains appropriate for Microsoft-native configuration, workflow, or rapid business application needs;
- configurable programme/project metadata;
- explicit beneficiary master-data model;
- generic monitored-entity and indicator models for non-beneficiary use cases;
- separate transactional submission records;
- normalized reference data;
- integration services for approved internal and external systems;
- analytics-ready reporting layer;
- secure identity and access management;
- monitored deployment environments for development, testing, staging, and production;
- backup, recovery, observability, and release management.

The final enterprise stack should be selected through architecture review, not assumed from the prototype. The documentation should therefore present Power Pages + Dataverse as the prototype/current architecture and the dedicated application/API/DBMS model as the scalable product direction.

## Analytics Direction

The prototype can provide simple portal KPI visualisation directly on Power Pages.

The future product should introduce a governed analytics layer, such as Power BI, when permissions, licensing, data governance, and ownership are approved.

Future analytics should support:

- programme performance dashboards;
- geographic breakdowns;
- intervention outcome analysis;
- baseline-to-endline comparisons;
- inclusion indicators;
- operational efficiency and environmental indicators;
- exportable reports for management and partners.

## Production Readiness Requirements

Before production rollout, the product should have:

- approved data protection and privacy review;
- confirmed access control model;
- validated data model and migration plan;
- confirmed target technology stack and hosting model;
- database administration, backup, recovery, and retention plan;
- user acceptance testing;
- administrator training;
- documented incident and support process;
- backup and recovery approach;
- deployment checklist and rollback approach.

## Roadmap Summary

| Stage | Focus | Outcome |
| --- | --- | --- |
| Prototype | TACATDP baseline collection, beneficiary-linked records, portal KPIs | Demonstrates feasibility and user value |

| Stage | Focus | Outcome |
|---|---|---|
| Pilot | Data quality, user feedback, configurable project model, refined dashboards, controlled rollout | Validates operational fit |
| Architecture review | Decide target frontend, backend API, DBMS/data platform, identity, analytics, and hosting model | Prevents prototype constraints from becoming production constraints |
| Production MVP | Security, governance, analytics, support, deployment process, selected enterprise stack | Supports real programme/project use |
| Scale-up | Integrations, advanced analytics, automation, multi-programme support, operations hardening | Enterprise MEL platform |